Graphing Basic Polar Curves from Equations

Answer Key

Precalculus

Quick Answers

1. 3	5. -2	9. 120, 240
2. 2	6. 180	10. 0
3. 1	7. 30, 150, 270	11. —
4. 0, 90, 180, 270	8. 4	12. —

Curriculum

Level: Precalculus
HSF-TF.A.2, HSF-TF.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

5. *If they got 6: took cos of 180 degrees as +1 instead of -1, so the negative r never appeared*

1. $r = 6\cos\theta$ at $\theta = 60°$. Substitute the angle and use the exact cosine value: $\cos 60° = \frac{1}{2}$, so $r = 6 \cdot \frac{1}{2}$. $\boxed{r = 3}$

2. $r = 4\sin\theta$ at $\theta = 30°$. Since $\sin 30° = \frac{1}{2}$, $r = 4 \cdot \frac{1}{2}$. $\boxed{r = 2}$

3. $r = 1 + \cos\theta$ at $\theta = 90°$. Here $\cos 90° = 0$, so $r = 1 + 0$. The cardioid therefore reaches one unit straight up the $90°$ ray. $\boxed{r = 1}$

4. A rose returns to the pole exactly where $r = 0$:

$$4\sin(2\theta) = 0 \quad\Longrightarrow\quad \sin(2\theta) = 0$$

Sine vanishes at multiples of $180°$, so $2\theta = 0°, 180°, 360°, 540°$ — the doubled window is what produces four answers, not two. Halving each gives

$$\boxed{\theta = 0°,\ 90°,\ 180°,\ 270°}$$

Between consecutive zeros the curve traces one petal, which is why $r = 4\sin(2\theta)$ has four petals rather than two.

5. $r = 2 + 4\cos\theta$ at $\theta = 180°$. Since $\cos 180° = -1$, $r = 2 + 4(-1) = 2 - 4$. $\boxed{r = -2}$
A negative r is not an error: the point is plotted 2 units from the pole in the direction *opposite* 180°, i.e. along the 0° ray. A student who takes $\cos 180°$ as $+1$ gets 6 and misses the inner loop entirely.

6. Set the cardioid's radius to zero:

$$1 + \cos\theta = 0 \quad\Longrightarrow\quad \cos\theta = -1$$

On $0° \le \theta < 360°$ the cosine equals -1 at exactly one angle. $\boxed{\theta = 180°}$ This lone zero is the cardioid's cusp, which is what distinguishes a cardioid from a limacon with an inner loop (two zeros) or one without (none).

7. The petal tips are where the radius attains its maximum 2:

$$2\sin(3\theta) = 2 \quad\Longrightarrow\quad \sin(3\theta) = 1$$

Sine equals 1 at 90° plus multiples of 360°, so $3\theta = 90°,\ 450°,\ 810°$ (the next value, 1170°, would put θ outside the window). Dividing by 3:

$$\boxed{\theta = 30°,\ 150°,\ 270°}$$

Three tips, 120° apart — the three-petal rose you expect from an odd coefficient.

8. $r = 3 - 2\cos\theta$ at $\theta = 120°$. Since $\cos 120° = -\frac{1}{2}$, $r = 3 - 2\left(-\frac{1}{2}\right) = 3 + 1$. Subtracting a negative increases r, which is why this limacon bulges toward 180°. $\boxed{r = 4}$

9. The inner loop begins and ends where the curve passes through the pole:

$$2 + 4\cos\theta = 0 \quad\Longrightarrow\quad \cos\theta = -\tfrac{1}{2}$$

Cosine is $-\frac{1}{2}$ in the second and third quadrants, at

$$\boxed{\theta = 120°,\ 240°}$$

Between those two angles r is negative, and those points plot on the opposite side of the pole — that is the inner loop.

10. The circle's maximum radius occurs where

$$4\cos\theta = 4 \quad\Longrightarrow\quad \cos\theta = 1$$

On $0° \le \theta < 360°$ this happens once. $\boxed{\theta = 0°}$
So $r = 4\cos\theta$ is the circle of diameter 4 sitting on the polar axis, with centre $(2, 0°)$ — the maximum-r direction is exactly the direction of the offset.

11. Manual review — hand-drawn sketch. A correct graph shows:

- The four quadrantal points: $r = 2$ at $\theta = 0°$, $r = 1$ at 90°, $r = 0$ at 180°, $r = 1$ at 270°.
- A smooth heart shape symmetric about the horizontal polar axis (because cos is even, $r(-\theta) = r(\theta)$).
- The cusp labelled at the pole, reached along the 180° direction.
- No part of the curve outside the ring $r = 2$, and no negative r: the radius $1+\cos\theta$ never drops below 0.

12. Manual review — identification, sketch, and justification. A complete answer contains:

- *Family:* a limacon of the form $r = a + b\cos\theta$, specifically a limacon with an inner loop.
- *Test:* compare a with b. Here $a = 2$ and $b = 4$, so $a/b = 1/2 < 1$; the radius must change sign, and a polar curve crosses the pole exactly where it does. For $r = 4+2\cos\theta$ the ratio is $2 > 1$, the radius stays positive, and no loop can form.
- *Evidence:* the pole crossings found in problem 9 are $\theta = 120°$ and 240°; between them $r < 0$ (for instance $r = -2$ at 180°, from problem 5), and those negative radii plot back through the pole to form the loop.
- *Sketch:* outer curve reaching $r = 6$ at $\theta = 0°$, passing through the pole at 120° and 240°, with the small loop lying on the 0° side of the pole and extending to a distance 2. Symmetric about the polar axis.

Accept any justification that compares the constant with the coefficient and links the sign change of r to the pole crossings.